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Loan Default Prediction Based on Logistic Regression and XGBoost Modeling

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***Abstract*—The accurate prediction of loan default risk is of paramount importance in the financial sector. In this paper, we delve into the realm of predictive modeling by employing logistic regression and XGBoost algorithms to forecast loan default occurrences. Our methodology involves rigorous data preprocessing, including handling missing values and encoding categorical variables, followed by the training of logistic regression and XGBoost models on the refined dataset. Our experimental findings reveal that the XGBoost model surpasses logistic regression in predictive accuracy. Furthermore, through comprehensive feature importance analysis, we identify and elucidate the key determinants contributing to loan default prediction. The insights garnered from our study hold significant implications for financial institutions, offering valuable guidance in the assessment and mitigation of loan default risks, thereby fostering a more secure and stable lending environment.**

***Keywords*—Loan default risk, Logistic regression, XGBoost algorithm**

I. INTRODUCTION

In 2022, the People's Bank of China released the "Opinions on Enhancing Financial Support for Comprehensive Rural Revitalization 2022." These guidelines aimed to steer the financial system towards optimizing resource allocation, implementing precise and forward-looking strategies, and bolstering financial support in the realm of the "Three Rural Areas." The objective was to sustain comprehensive rural revitalization and provide substantial backing to stabilize the overall macroeconomic situation. Lately, supply chain finance has been introduced into the agricultural sector both domestically and internationally. Agricultural supply chain finance integrates the information flow of upstream and downstream enterprises, with core agricultural enterprises at the center. Leveraging the credit advantage of these core enterprises, it employs guaranteed financing forms such as accounts receivable, prepaid accounts, and inventory to effectively address issues related to information deficiency and the lack of effective guarantees for small and medium-sized micro-enterprises. However, in the traditional supply chain finance model, the core enterprise releasing information holds a natural advantage, thereby directly impacting the operational effectiveness of supply chain finance. Consequently, the introduction of new technological means to overcome information barriers between the agricultural supply chain and financial service institutions, while simultaneously addressing the financing challenges of farmers, represents the current trend in agricultural research and practice.

Agricultural certificate loans benefit Lin'an farmers. The agricultural certificate loan product is an online financial product that serves farmers and agricultural enterprises based on the supply chain financial model of Farmers and Merchants Bank and created through blockchain technology, which is an agricultural supply chain financial product that applies blockchain technology.

Within the financial domain, the prediction of loan defaults has consistently been a topic of significant interest. As the financial market continually evolves and risk management grows in importance, precise forecasting of loan defaults has emerged as a pivotal undertaking for financial institutions and investors. By effectively predicting loan defaults, financial institutions can better assess risks, set reasonable loan interest rates and approval criteria, and thus minimize losses. Therefore, the research and application of loan default prediction models are of great significance to the financial industry.

This study aims to explore and compare the application of two models, Logistic regression and XGBoost, in loan default prediction. Logistic regression, as a classical statistical learning method, has a wider application in loan default prediction, whereas XGBoost, as an integrated learning algorithm based on decision trees, has achieved remarkable success in the field of data mining and predictive modeling in recent years. Through the comparison of the performance of these two models in loan default prediction, our goal is to furnish financial institutions with more precise and dependable prediction tools. These tools are designed to effectively mitigate risk and enhance the efficiency and accuracy of loan decisions.

In this study, we will use a bank's loan data as a sample, and construct and compare the prediction effects of Logistic Regression and XGBoost models through the steps of data cleaning, feature engineering, and model training. Through the empirical analysis in this study, we expect to provide certain reference and references for the selection and application of loan default prediction models, and provide more practical value for financial risk management and decision-making.

II. LITERATURE REVIEW

The study by Zhang et al. [1] introduced a novel profit-driven prediction model for optimizing the hyper-parameters of the predictor-classification boost using the profit metric as the objective for Bayesian optimization. Subsequently, Shapley additive explanatory (SHAP) values were computed to further elucidate the relationship between input variables and predicted values. Utilizing two datasets from Renren and Lending Club, the experimental results and statistical tests demonstrated that the proposed model achieved the highest values of profit-related evaluation metrics. It yielded an average additional profit margin of 3.0872% and 2.1858%, along with an average profit of 5,168.8762 and 352.9787, respectively. Furthermore, the SHAP values unveiled the key factors influencing the predicted output, offering valuable insights for platforms and lenders to identify potential defaulting parties. Owusu et al. [2] focused on addressing loan defaults within online peer-to-peer (P2P) lending activities. They obtained data for their study from online lending clubs on the Kaggle platform, with loan status selected as the dependent variable and discretized into "default" and "full" loans. The dataset underwent preprocessing to remove all irrelevant instances. Given the unbalanced nature of the dataset, the Adaptive Synthesis (ADASYN) oversampling algorithm was utilized to address data imbalance by oversampling a few classes. For prediction, a Deep Neural Network (DNN) was employed, achieving a prediction accuracy of 94.1%, the highest score obtained across several trials for batch size and epoch change.

In a different vein, Song et al. [3] acknowledged the varying risk preferences of lending companies during the review of loan applications. They proposed a multi-objective integrated learning framework for ratings, employing a modeling strategy tailored to credit ratings to construct candidate models for subpopulations of loans with similar default risks. The approach involved using an effective unbalanced classification method, a single-class support vector machine, as the base learning algorithm. An adaptive classification boundary adjustment process was proposed to enhance imbalance classification performance. To validate their method, the authors conducted a comprehensive experimental study and compared it with several benchmark algorithms using industry-provided datasets. The experimental results demonstrated the advantages of rating-specific strategies and the superiority of the proposed integrated multi-objective learning approach.

In their work, Chen et al. [4] focused on the function extraction and data processing of basic customer attribute data and transaction data, derived from a bank credit application scenario. Subsequently, they proposed a linear regression model with penalties, as well as a neural network prediction model, aimed at enhancing the accuracy of bankruptcy assessment and achieving local optimization. This approach sought to improve implicit risk prediction and control of customer credit, consequently leading to a notable reduction in the default risk of bank loans. By considering the characteristics of the collected sample data, the most suitable penalized linear regression prediction algorithm was selected and experimentally analyzed to enhance the bank's risk management. The experimental results demonstrated that the improved logistic regression and neural network models exhibited significant advantages over the predictive effects of all four models. In their study, Ma et al. [5] employed "multi-observation" and "multi-dimensional" data cleaning methods and utilized the 2016 end-of-year Asian Modern machine learning algorithms LightGBM and XGboost, based on real P2P transaction data from Lending Club. They presented a robust and innovative prediction of the platform's loan default risk, comparing the results of different methods. The paper noted that the LightGBM algorithm, which categorized the prediction results based on multiple observation datasets, yielded the best performance. Specifically, the average performance rate of the Lending Club platform's historical transaction data increased by 1.28 percentage points, leading to a reduction in loan defaults by approximately $117 million. Finally, the paper offered suggestions for the development of P2P platforms such as Lending Club and others, outlining factors affecting default rates in these platforms. In their work, Liu et al. [6] utilized an enhanced machine learning model fusion algorithm to further enhance the recall and AUC metrics for P2P credit default prediction, employing Lending Club's lending dataset and focusing on the ROC curve's area under the curve (AUC).

III. LOAN DEFAULT PREDICTION

Methods in statistics include linear regression, logistic regression, etc. While considering loan default as a categorical problem, the variable components are more diversified and there is no fixed pattern association between samples and variables. If accuracy is the goal, XGBoost is preferred by considering factors such as robustness, speed, and versatility, etc. Since some banks currently use an expert scoring scale combined with a logistic regression model to measure whether a user has defaulted or not, this paper uses logistic regression. In addition, they used XGBoost to compare the prediction effectiveness of the two models. The modeling and solution idea of this paper is shown in Figure 1.

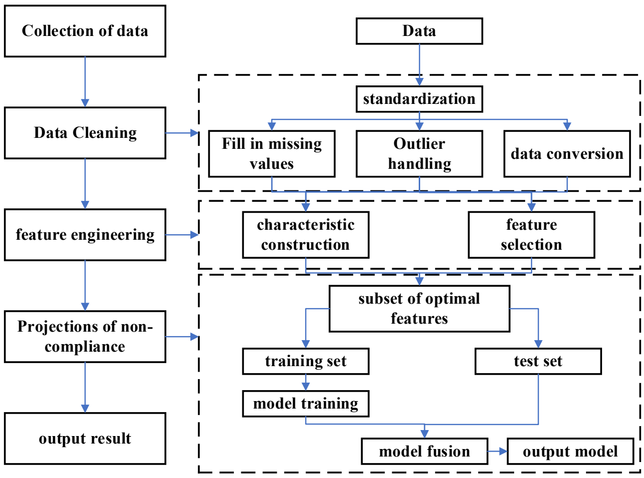

Fig. 1. Modeling and Solution Ideas

Default prediction of borrowers is essentially a binary classification problem, where x denotes the basic information and historical behavioral data of the user, and the predictor variable y is the expected loan profile of each borrower, denoted by sigmoid coded values:

$$Sigmoid = \begin{cases} 0, \text{normal repayment} \\ 1, \text{default} \end{cases} \quad (1)$$

The output of the final model is also 0, 1 value, which represents the customer default or not. It's crucial to address strong correlations among indicators to enhance the training effectiveness. To achieve this, the elimination of correlated factors is necessary, enabling the selection of the most illustrative indicators. After eliminating redundant indicators, the final six variables are chosen as inputs to the model. These variables include:

1)Loan grade (grade)

2)Status of home ownership provided by the borrower when registering

3)Annual income

4)Debt-to-income ratio (dti)

5)Number of defaults in the borrower's credit file over the past 2 years that were more than 30 days past due (delinquency_2years)

6)Number of derogatory public records (pubRec)

Additionally, there is a labeled column indicating whether or not the loan is in default (isDefault). The correlations between these variables are depicted in Figure 2.

	grade	homeownership	annualincome	dti	delinquency_2years	pubRec
grade	1.00	0.17	0.18	0.46	0.29	0.22
homeownership	0.17	1.00	0.89	0.74	0.30	0.05
annualincome	0.18	0.89	1.00	0.58	0.40	0.22
dti	0.46	0.74	0.58	1.00	0.58	0.45
delinquency_2years	0.29	0.30	0.40	0.58	1.00	0.19
pubRec	0.22	0.05	0.22	0.45	0.19	1.00

Fig. 2. Correlation coefficients between the six input values

After cleaning and feature engineering of customer loan data, the appropriate model is selected for learning and the final evaluation metric is the AUC value.

A. Logistic Regression

Suppose an applicant's default rate P is related to the feature vector $X = (X_1, X_2, \ldots, X_P)$ has the following relationship:

$$P = W_0 + W_1X_1 + W_2X_2 + \ldots + W_pX_p + \varepsilon \tag{2}$$

where ε is a random perturbation term that determines W_p from the sample data and thus assesses the applicant's default rate after taking logarithms on both sides of Eq:

$$log\left(\frac{p_i}{1-p_i}\right) = W_0 + W_1X_1 + W_2X_2 + \ldots + W_pX_p = Wx^T \tag{3}$$

Since $\frac{p_i}{1-p_i}$ takes values from 0 to ∞ and $log\left(\frac{p_i}{1-p_i}\right)$ takes values from -∞ to +∞, both sides of the equation are then taken as logarithms with *e* as the base to obtain the logistic regression hypothesis:

$$p_i = \frac{e^W x}{1 + e^W x} \tag{4}$$

The six categories of farmers' data are entered into the model to obtain the predicted default rates of farmers, and Table I shows the predicted results for the 12 A-E-rated farmers.

TABLE I. LOGISTIC MODEL PREDICTS THE DEFAULT RATE OF FARMERS

ID	grade	subGrade	homeOwnership	annualIncome	dti	delinquency_2years	pubRec	isDefault	default rate
1	E	E2	2	110000	17.05	0	0	1	0.89
2	D	D2	0	46000	27.83	0	0	0	0.33
3	D	D3	0	74000	22.77	0	0	0	0.18
4	A	A4	1	118000	17.21	0	0	0	0.25
5	C	C2	1	29000	32.16	0	0	0	0.14
6	A	A5	0	39000	17.14	0	0	0	0.20
7	A	A4	0	35000	17.49	0	0	0	0.22
8	C	C3	1	30000	32.6	0	1	0	0.05
9	C	C2	2	60000	19.22	0	0	1	0.93
10	B	B4	1	15300	24.39	0	0	0	0.17
11	A	A1	3	150000	54.1	0	0	0	0.08
12	C	C1	1	100000	13.46	1	1	1	0.97

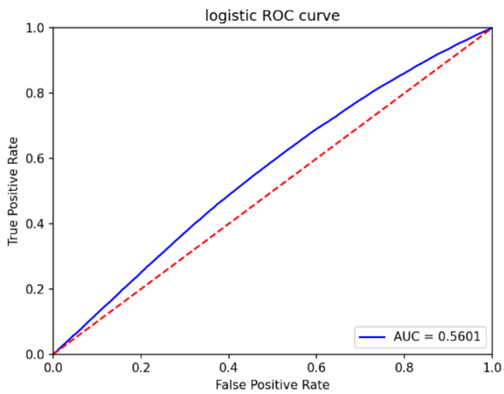

Fig. 3. ROC plot for the logistic model

The horizontal axis of the ROC curve represents the false positive rate, while the vertical axis represents the true positive rate. The AUC (Area Under the Curve) is defined as the area below the ROC curve enclosed by the axes. A larger AUC indicates a higher likelihood that the positive samples will be ranked at the forefront, implying superior classification.

Continuing to utilize the AUC value to assess the model's accuracy, Figure 3 illustrates the ROC curve of the logistic model along with the AUC value.

B. XGBoost model

XGBoost objective function:

$$L(\theta) = \sum_{i}^{n} l(y_i, y^{(t-1)}) + f_{t(x_i)} + \Omega(f_t) \tag{5}$$

The second-order Taylor expansion of the objective function is organized to obtain the optimal solution:

$$L(\theta) = -\frac{1}{2}\sum_{J=1}^{T}\frac{\sum g_i}{\sum h_i} + \gamma T_I \tag{6}$$

Gradient ascent is utilized to find the globally optimal parameters.

In the construction process of a tree model for a given dataset, a feature segmentation point is greedily chosen at each layer to serve as a leaf node, aiming to maximize the overall gain of the tree after segmentation. This implies that the more frequently a feature is segmented, the more advantageous it is for the entire tree, indicating the feature's importance. Similarly, the greater the average gain of a feature each time it is segmented, the more significant the feature becomes.

During the segmentation process, the weight of each leaf node can be expressed as $w(gi, hi)$, where gi and hi represent specific parameters.

$$g_i = \partial_{\hat{y}^{(t-1)}} l\left(y_i, \hat{y}^{(t-1)}\right) \tag{7}$$

$$h_i = \partial^2_{\hat{y}^{(t-1)}} l\left(y_i, \hat{y}^{(t-1)}\right) \tag{8}$$

The training error $\left(y_i, \hat{y}\right)$ signifies the difference between the target value y_i and the predicted value $\hat{y}$. To minimize the cost of the segmented tree, the gain (Gain) of each feature as a segmentation point is evaluated based on the weights of all the leaf nodes, expressed as:

$$\text{Gain} = \sum_{\text{left}} w + \sum_{\text{right}} w - \sum_{\text{nosplit}} w \tag{9}$$

The decision tree model, as a non-parametric supervised learning model, is frequently employed for both classification and regression tasks. This model doesn't rely on any a priori assumptions about the data and can rapidly derive decision rules based on the data's characteristics.

XGBoost adopts an integration strategy based on decision trees, utilizing the gradient boosting algorithm to iteratively minimize the loss of previously generated decision trees and create new trees to build the model, thereby ensuring the reliability of the final decision. XGBoost adds one tree at each iteration and constructs a linear combination of K trees as:

$$\hat{y}_i^{(t)} = \sum_{k=1}^{K} f_k(x_i) = \hat{y}_i^{(t-1)} + f_t(x_i), f_k \in F \tag{10}$$

Here, F represents the function space containing all trees, and $f_t\left(x_i\right)$ denotes the weight of the i-th sample being classified to the leaf node where it is located in the k-th tree.

For the above three important metrics, there are:

$$\text{FScore} \,\&= |X| \tag{11}$$

$$\text{AverageGain} \,\&= \frac{\sum \text{Gain}_X}{\text{FScore}} \tag{12}$$

$$\text{AverageCover} \,\&= \frac{\sum \text{Cover}_X}{\text{FScore}} \tag{13}$$

For the above three important metrics, there are:

X: the set of desired features classified to the leaf nodes

Gain: the node gain value at the time of segmentation for each leaf node in X obtained from the above equation

Cover: the number of samples falling on each node in X

The XGBoost model training optimization parameters are shown in Table II.

TABLE II. OPTIMIZATION PARAMETER

Parameter name	Parameter Meaning	value
gamma	Number of null leaves	0
Max_depth	Maximum depth per tree	5
Min_child_weight	The minimum weight of each leaf	1

TABLE III. EXAMPLES OF MODEL INPUT PARAMETER VALUES

ID	grade	subGrade	homeOwnership	annualIncome	dti	delinquency_2years	pubRec	isDefault	default rate
1	E	E2	2	110000	17.05	0	0	1	0.99
2	D	D2	0	46000	27.83	0	0	0	0.23
3	D	D3	0	74000	22.77	0	0	0	0.12
4	A	A4	1	118000	17.21	0	0	0	0.25
5	C	C2	1	29000	32.16	0	0	0	0.11
6	A	A5	0	39000	17.14	0	0	0	0.21
7	A	A4	0	35000	17.49	0	0	0	0.12
8	C	C3	1	30000	32.6	0	1	0	0.09
9	C	C2	2	60000	19.22	0	0	1	0.95
10	B	B4	1	15300	24.39	0	0	0	0.07
11	A	A1	3	150000	54.1	0	0	0	0.01
12	C	C1	1	100000	13.46	1	1	1	0.98

The data were entered into the credit scoring system and the predictions of the XGBoost model were obtained, as shown in table III.

From the ROC plot with the AUC values, it is evident that the AUC of XGBoost is greater than the AUC of the logistic model regression, signifying that the XGBoost model is more accurate, as illustrated in Figure 4.

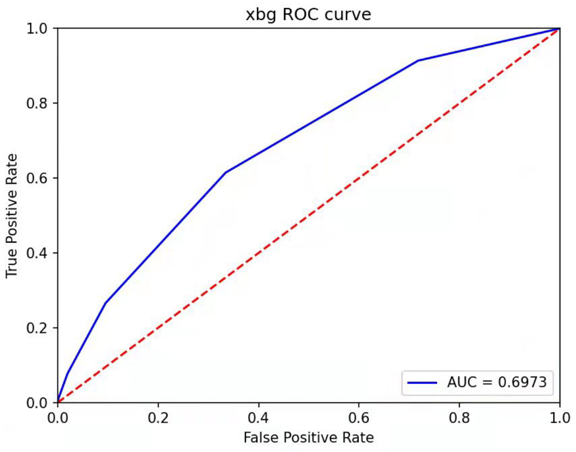

Fig. 4. ROC plot for XGBoost model regression

IV. Conclusion

In this study, we have meticulously examined the efficacy of logistic regression and XGBoost models in the context of loan default prediction. Our comprehensive analysis demonstrates the superiority of the XGBoost model in terms of predictive performance when compared to logistic regression. Moreover, the in-depth feature importance analysis has shed light on the pivotal factors influencing loan default occurrences, thereby providing actionable insights for risk assessment and management. The implications of our findings underscore the pivotal role of advanced machine learning techniques in fortifying the accuracy of loan default prediction, which can significantly benefit financial institutions in formulating robust risk management strategies. Looking ahead, further research endeavors could focus on the exploration of additional features and the application of more sophisticated ensemble learning techniques to further enhance predictive capabilities and bolster the resilience of lending practices.

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